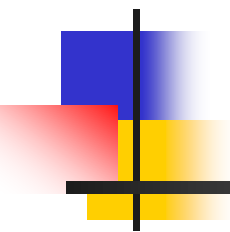


Marine microbiology

Winter Semester

Lecture 3



Tools for the study of marine microbial diversity

References:

Munn. Marine Microbiology (MM): Chapter 2

Prescott's Microbiology (PM): Chapters 2, 29

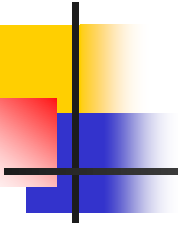
第十章 MOOC 海洋微生物的研究技术(三节内容)



Objectives for this lecture:

At the end of this lecture, you should be able to:

1. Know the use of remote sensing in ocean monitoring
2. Know the use of biomarkers of microbial activity
3. Describe the main **culture-dependent** methods used in marine microbiology
 - Enrichment culture
 - Selective isolation, involving spread plating
 - Single cell isolation/laser tweezers
4. Describe the main **culture-independent** methods used in marine microbiology
 - Microscopy (light, fluorescence, EM, confocal, scanning probe)
 - Flow cytometry
 - Fluorescence *in situ* hybridization (FISH)
 - Polymerase chain reaction (PCR) for community analysis
 - Environmental genomics (metagenomics) – next lecture



Introduction

- The underlying principles of methods need to be understood in an introductory course, but not the technical details
- Traditional methods based on culture and microscopy are always essential for marine microbiology
- New techniques in marine microbiology have resulted in a huge recent increase in our knowledge of marine microbes

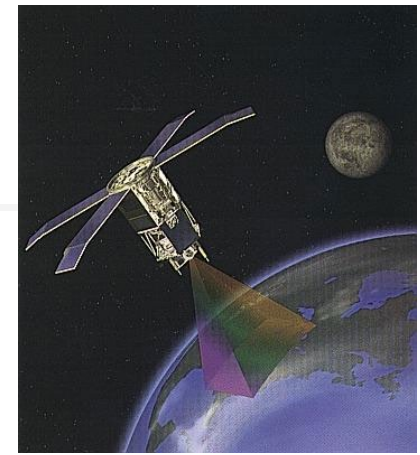
There are two fundamental questions in **marine microbiology**:

Who's there?

What are they doing?

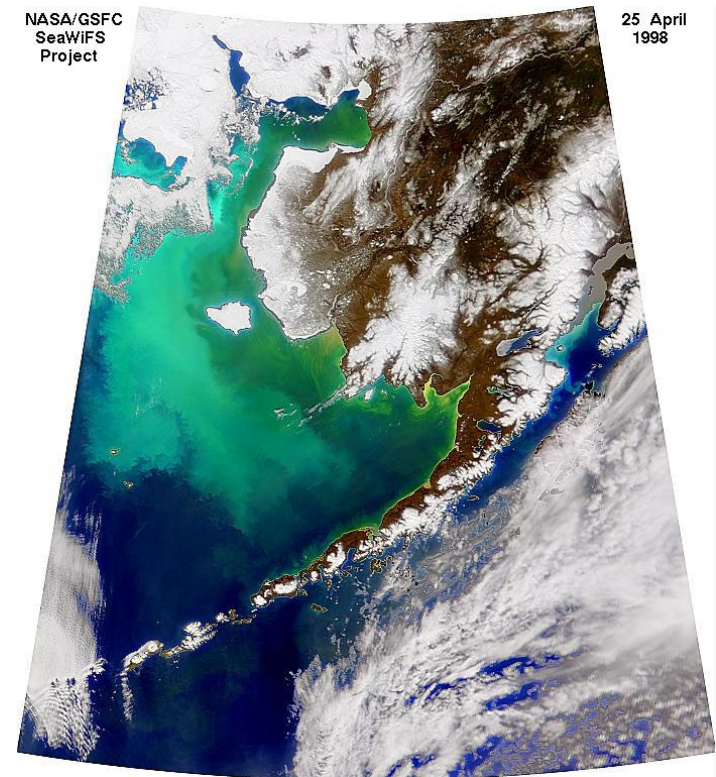
1. The big picture: remote sensing

遥感技术



- **Remote sensing** has provided valuable information about the activities of planktonic organisms in **surface** layers of oceans
- The first was [Coastal one color scanner \(CZCS\)](#) (海岸带水色扫描仪) launched in 1978, ceased in 1986
 - collected light reflected from **surface** at different wavelengths including absorption max. for chlorophyll 683 nm
 - provided information about **distribution of primary producers**
- Scanning then carried out by [Sea-viewing Wide Field-of-View Sensor \(SeaWiFS\)](#) 海洋观测广角感测仪
 - used to monitor phytoplankton world-wide, including algal blooms

A true color SeaWiFS image of a
coccolithophore phytoplankton
bloom off Alaska



1. The big picture: remote sensing

- Currently a number of 3rd-generation satellite sensors from various countries (China ~10, USA, Europe, Japan, India etc.)
- Different phytoplankton species have different number and type of spectral peaks



Surface accumulations of cyanobacteria *Nodularia* in the Baltic Sea as detected by enhanced "true-color" MODIS imagery on July 30, 2003.

栏目概况

[首页](#) > [数据集](#) > [栏目概况](#) >

栏目概况



SIO地面站实时遥感
产品



海洋碳汇数据集



中国海水质数据集



全球生态环境数据集



东中国海生态环境逐
时数据集

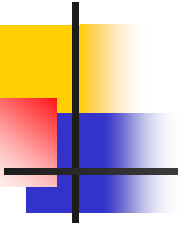


全球实测数据网格化
产品



全球海洋再分析数据
集

 SIO地面站实时遥感产品 海洋碳汇数据集 中国海水质数据集 全球生态环境数据集 东中国海生态环境逐时数据集 全球实测数据网格化产品 全球海洋再分析数据集



2. General experimental procedures

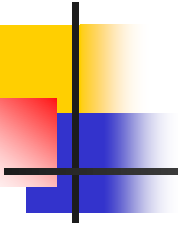
Planktonic communities are difficult to study because of:

1. **The small sizes** of the individuals in the populations involved
2. **The absence of simple tools** applicable to all functional and taxonomic groups
3. **The enormous** genetic and taxonomic **diversity** involved

Measuring techniques must be designed to study organisms

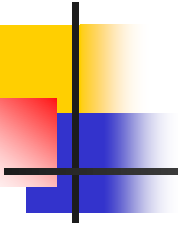
in situ = in place (原位监测)

- Microbes may exist in a 3D structured and diverse organization
e.g. anaerobic, microaerophilic and aerobic conditions may all occur in a sediment particle or a microbial mat
- Bulk measurements of substrates, pH, oxygen etc. may not reflect microenvironments



2. General experimental procedures

- Measurement of specific cell constituents may be used as biomarkers of microbial activity
 - **ATP** – occurs in all living organisms, indicator of live cells
 - **LPS (lipopolysaccharide, 脂多糖)** – indicator of Gram negative bacteria
 - ATP and LPS measurements common in 1980s but less widely used since molecular methods developed
 - **Photosynthetic pigments** – indicator of phytoplankton structure (Chlorophylls, bacteriophylls, carotenoids, and other pigments).
 - **Unique archaeal lipids** – indicator of archaea



3. Culture-dependent analyses

Culture of microbes depends on **the right combination of nutrients and growth conditions** e.g.

1. **Heterotrophic bacteria** – dissolved organic and inorganic compounds
2. **Phototrophs** – inorganic nutrients and light
3. **Bacteriophages** – susceptible bacterial host cells
4. **Phagotrophic protists** (吞噬型的原生生物) – suitable host prey cells

Techniques to be discussed:

- **Enrichment culture** 富集培养
- **Selective isolation, involving spread plating** 筛选分离
- **Single cell isolation** 单细胞分离技术

Enrichment culture 富集培养

The main approach to culturing *functional groups* (*guilds*, 同资源种团)

- Often used as the first step in culturing specific members (termed 'isolation') of that guild

Used in studies of marine bacteria:

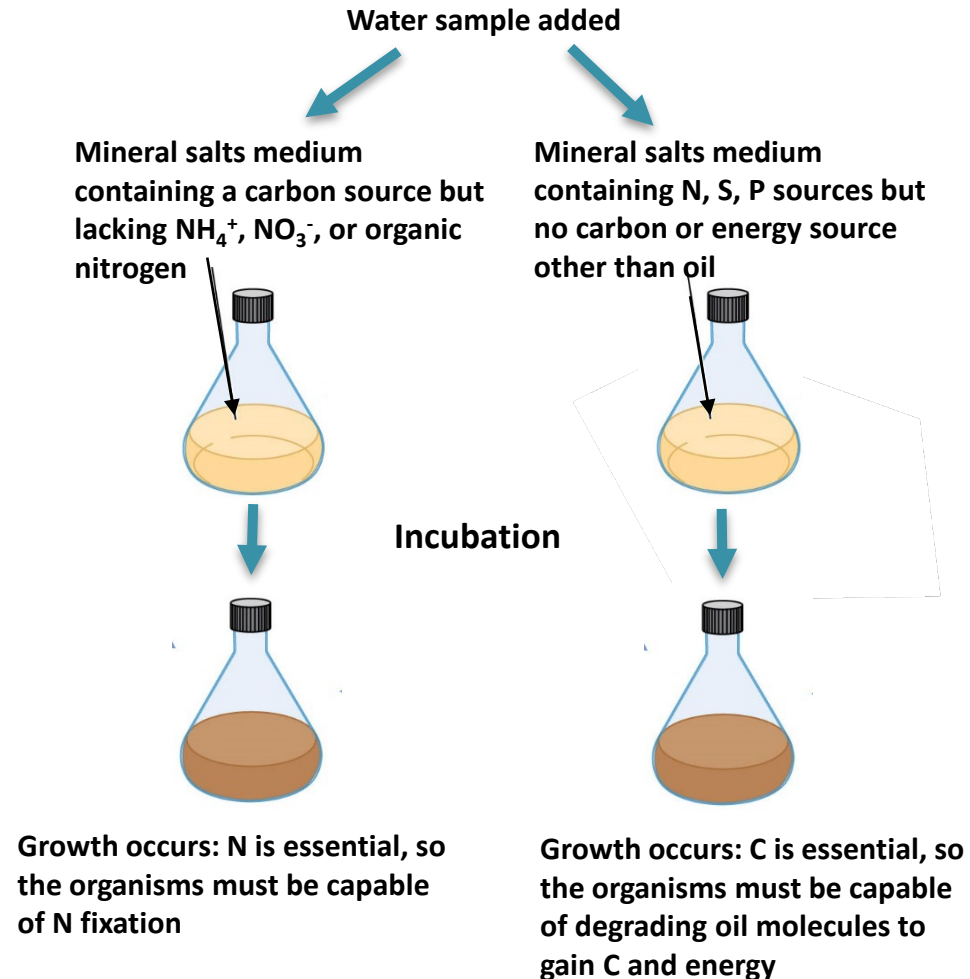
- fixing atmospheric N (N_2 fixers)
- degrading oil spills

Principle:

Conditions are imposed that selectively enrich for (increase the abundance of) members of the community having the activity of interest

Enrichment strategy for N_2 fixing microbes

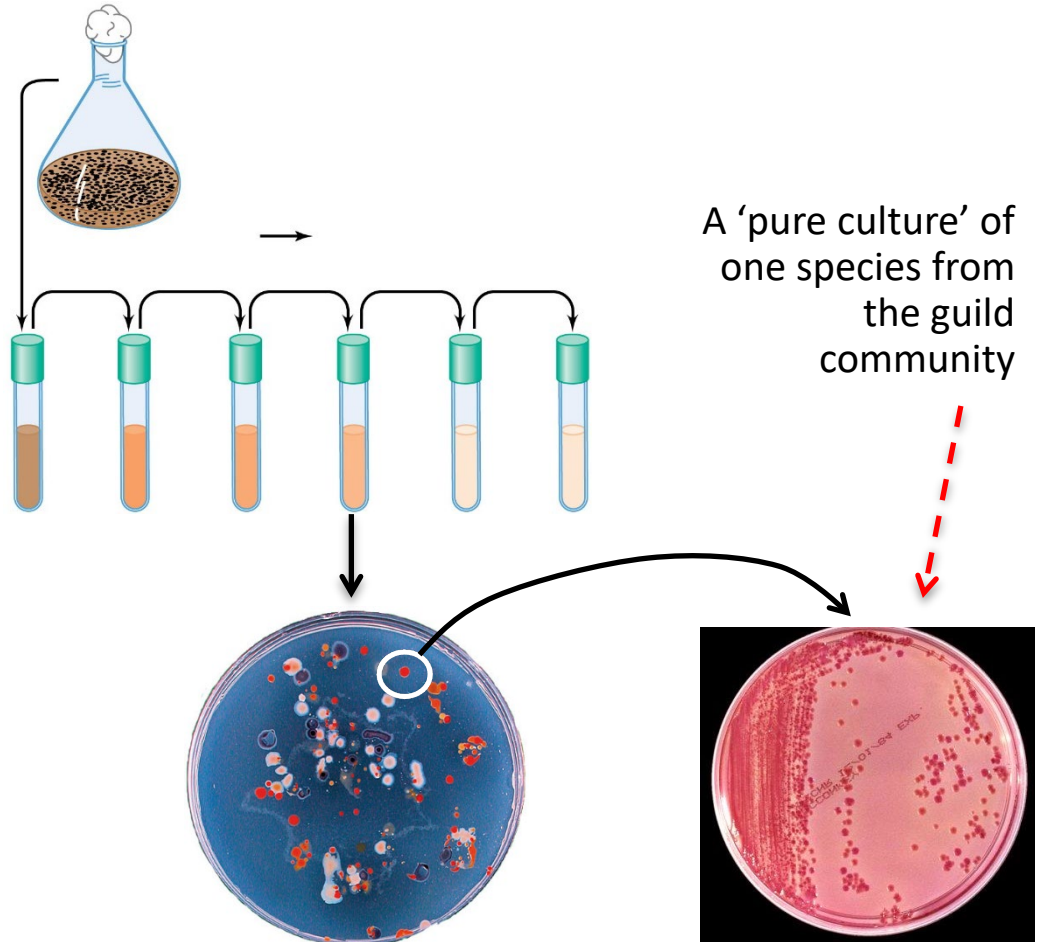
Enrichment strategy for oil-degrading microbes



Isolation of guild members by spread plating

Many studies of microbial oceanography have a focus on the characteristics of members of particular guilds

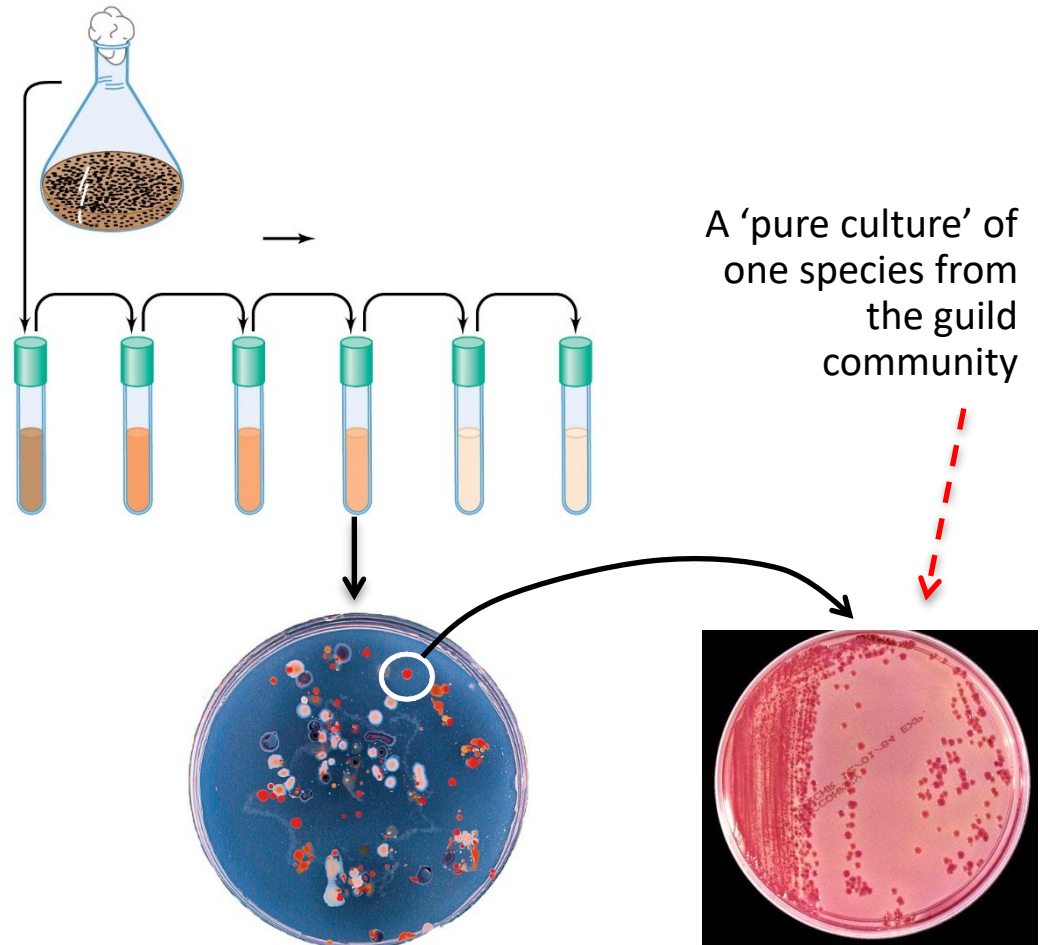
- This requires that representatives of the populations forming **the guild community are separated from each other and cultured**
- This can be achieved by a **combination of serial dilution and isolation culture**



Isolation of guild members by spread plating

Principle:

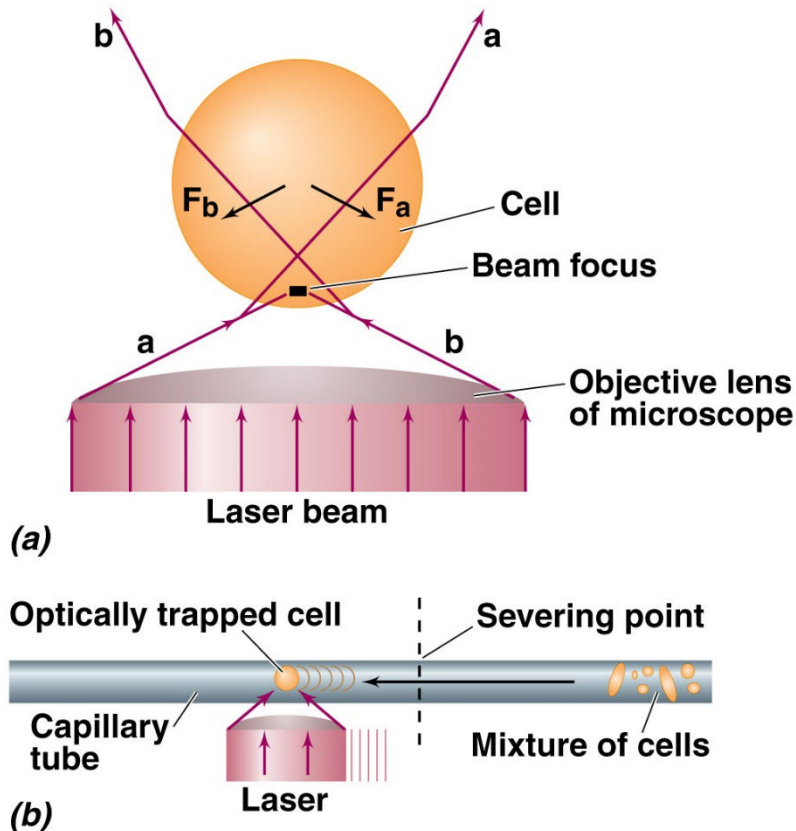
- **sample** or **enrichment** culture is serially diluted in sterile saline
- subsamples are transferred from each dilution to growth medium prepared in agar plates (e.g. the medium used for enriching N_2 fixers)
- at some dilution, separation of the various colony types is achieved
 - these represent the diversity of species in the sample
 - representatives of each species can be picked and transferred to agar medium free of other community members



Overcoming past limitations to culturing

It is now possible to select and separate *individual cells* from natural samples

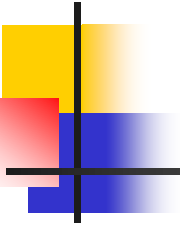
- a) useful for obtaining cultures of **slow growing species** in a guild
 - b) useful for culturing of species that constitute a very **small proportion** of the community
- in both cases above these members of the community would be missed in the enrichment and serial dilution processes discussed so far



Single-cell isolation using **laser tweezers**

Principle:

- A single cell in a mixed sample in a capillary tube is identified by microscopy
- The laser beam creates a pressure that holds the cell in place
- The beam (holding the cell of interest) is moved away from the other community members
- When sufficiently separated the capillary is severed (cut) and the trapped cell washed out into sterile growth medium.



4. Culture-independent analyses

- The development of microscopy and molecular techniques have revolutionized studies of marine microbial oceanography
- These techniques have the major advantage of enabling abundance and diversity data to be rapidly acquired without the need to culture community members

Techniques to be discussed:

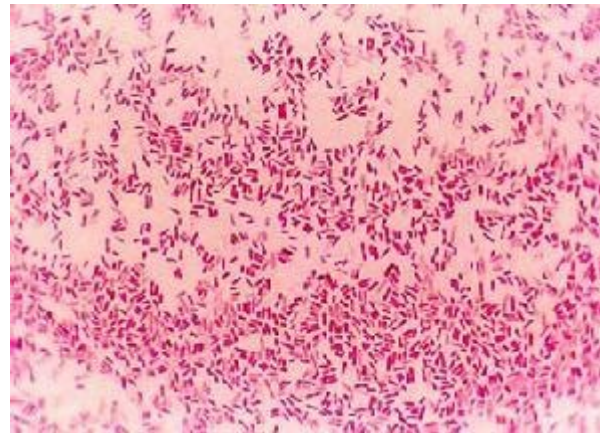
- Microscopy
- Flow cytometry
- Fluorescence *in situ* hybridization (FISH)
- Polymerase chain reaction (PCR) for community analysis
- Environmental genomics (metagenomics) – next lecture

Different types of Microscopes

Bright-Field Microscope 明视野显微镜



- It is called a bright-field microscope because it forms a dark image against a brighter background



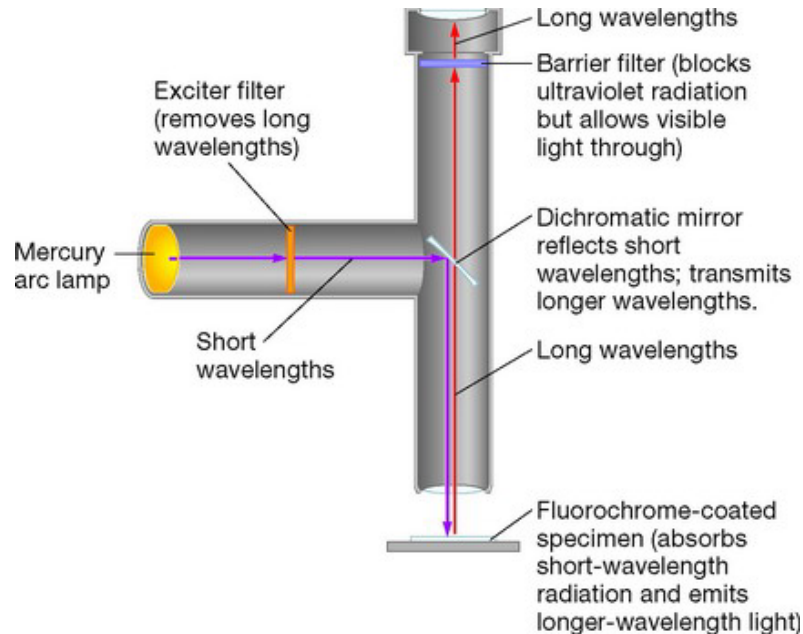
E.coli Germ stain, with oil immersion.

- Routinely used to examine both stained and unstained specimens (样本) in microbiology labs.

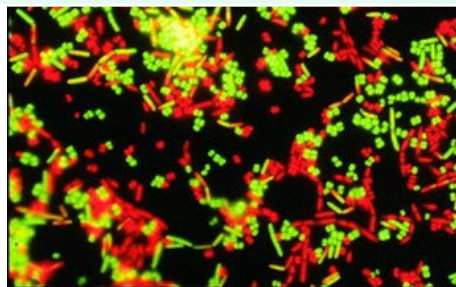
Different types of Microscopes

Fluorescence Microscope 荧光显微镜

- Exposes a specimen to **ultraviolet, violet or blue light** and forms an image of the object with the resulting fluorescent light.

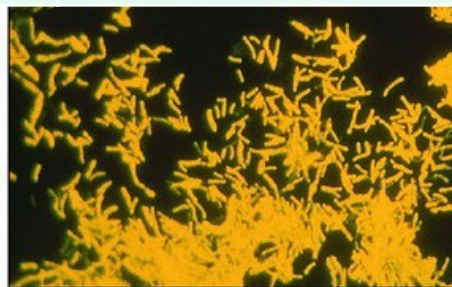


Fluorochrome (荧光染料)	Uses
Acridine orange (吖啶橙)	Stains DNA; fluoresces green (dsDNA) or red (ssDNA or RNA) https://www.thermofisher.com/order/catalog/product/A1301#/A1301
Diamidino-2-phenyl indole (DAPI)	Stains DNA; fluoresces blue to green https://www.thermofisher.com/cn/zh/home/life-science/cell-analysis/fluorophores/dapi-stain.html?SID=fr-dapi-main
Fluorescein diacetate (FDA) *	Accumulates in living cells; fluoresces green
Propidium Iodide (PI)	Stains DNA of dead cells; fluoresces red
Fluorescein isothiocyanate (FITC)	Often attached to antibodies that bind specific cellular components or to DNA probes; fluoresces green
Tetramethyl rhodamine isothiocyanate (TRITC or rhodamine)	Often attached to antibodies that bind specific cellular components; fluoresces red



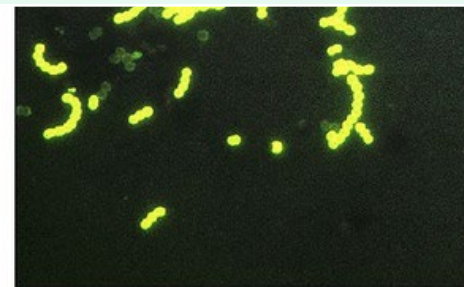
(a) 10 μm

Dyes that cause live cells to fluoresce green and dead ones red



(b) 10 μm

Auramine (金胺) is used to stain *Mycobacterium* 分支杆菌 species



(c) 10 μm

Fluorescent antibodies tag specific molecules.

Newer Techniques In Microscopy

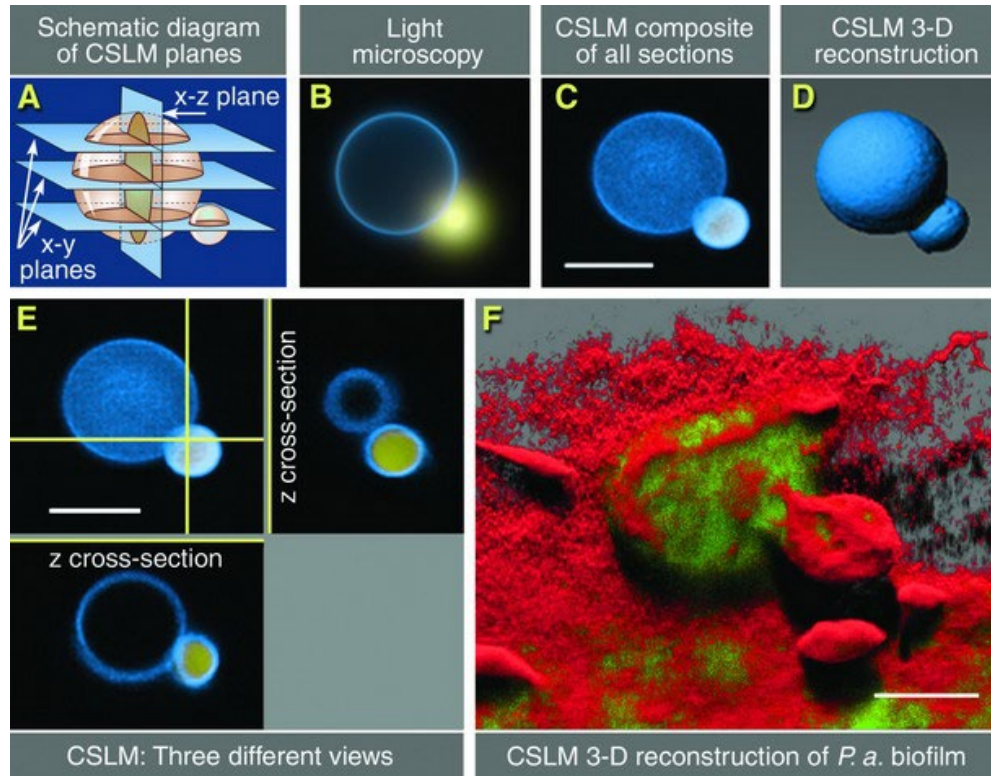
Confocal laser scanning microscope, Confocal scanning laser microscope ,
Confocal microscope

激光共聚焦显微镜

B. The light
microscope, just
one plane

C. A computer
collected from
multiple planes

A. The planes
observable by
confocal
microscope



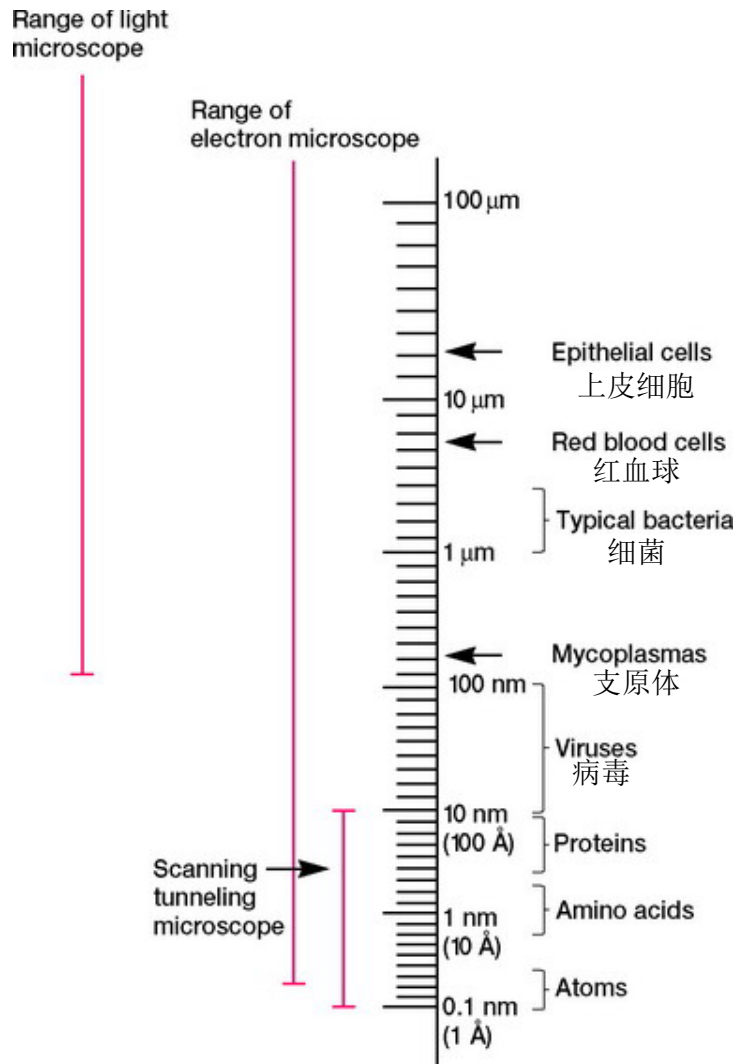
D. CSLM collected
multiple planes
and generated 3-
dimensional of the
specimen

F. A three-
dimensional
reconstruction of
a *Pseudomonas
aeruginosa* (铜绿假
单胞菌) biofilm

E. Computer
generated views of
a specimen

Different types of Microscopes

Electron Microscope 电子显微镜



The best **light microscopes** have a resolution limit of about **0.2 μm** , which greatly compromises their usefulness for detailed studies of many microorganism.

✗ Viruses——too small

✗ Internal structure of larger microbes

The reason is from the nature of visible light waves

Electron Microscopes

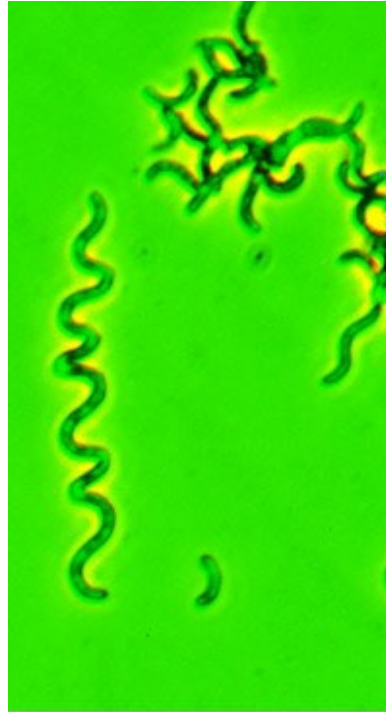
have much greater resolution.

➤ Transmission Electron Microscope (TEM) 透射电子显微镜

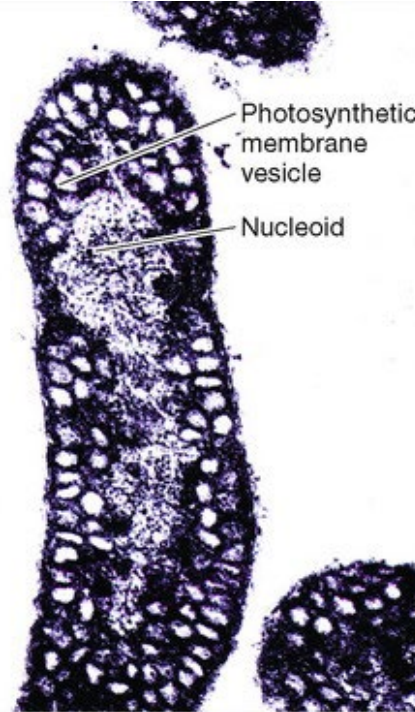
➤ Scanning Electron Microscope (SEM) 扫描电子显微镜

Different types of Microscopes

A comparison of light and Electron Microscopic resolution
(*Rhodospirillum rubrum*, 红螺菌)



Phase-contrast light microscope ($\times 600$)



A thin section of *R. rubrum* in TEM* ($\times 100,000$)

* TEM Transmission Electron Microscope

Different types of Microscopes

Transmission Electron Microscope 透射电子显微镜



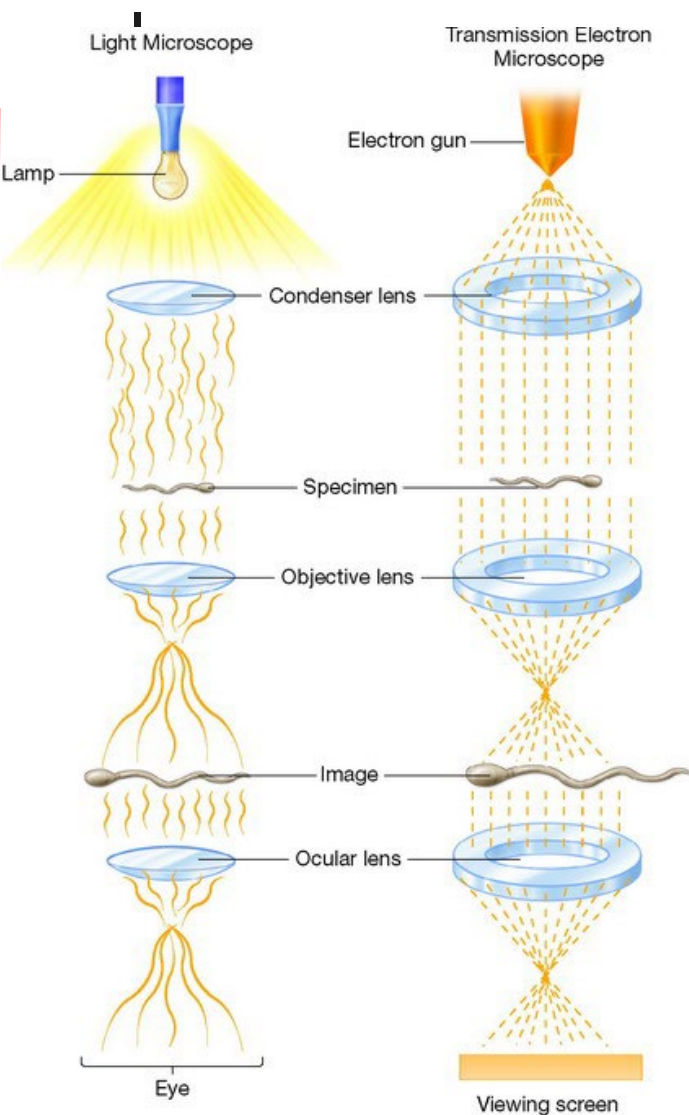
Final image can be displayed on fluorescent screen or photographed.

Electron gun

Specimen holder

Fluorescent screen

- **Resolution** of a light microscope **increases** with a **decrease** in the **wavelength** of the light for illumination
- Use **a beam of electrons** to illuminate and create magnified images of specimens
- Resolution of EM (0.2 nm) roughly 1,000 times better than the light microscope (0.2 μm)
- The useful magnification is well over 100,000 $\times$



Feature	Light Microscope	TEM
Radiation source	Visible light	Electron beam
Types of Lens	Glass	Electromagnet 电磁场
Specimen	Glass slide	Metal grid
Medium of travel	Air	High vacuum
Magnification	1,000-1,500	Over 100,000
Best resolution	0.2 μm	0.2 nm

* The resolution limit of a human eye is about 0.2 mm

Specimen preparation techniques:

➤ Negatively staining

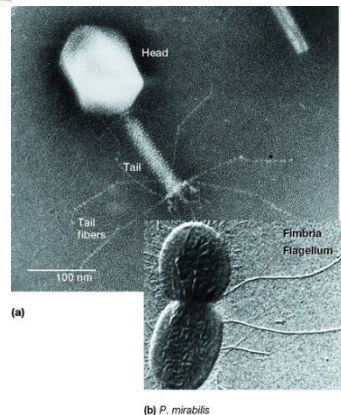
heavy metal render the background dark, specimen bright;

Excellent for viruses, bacterial gas vacuoles, etc.

➤ Shadowing

specimen is coated with heavy metal, the area coated with metal dark

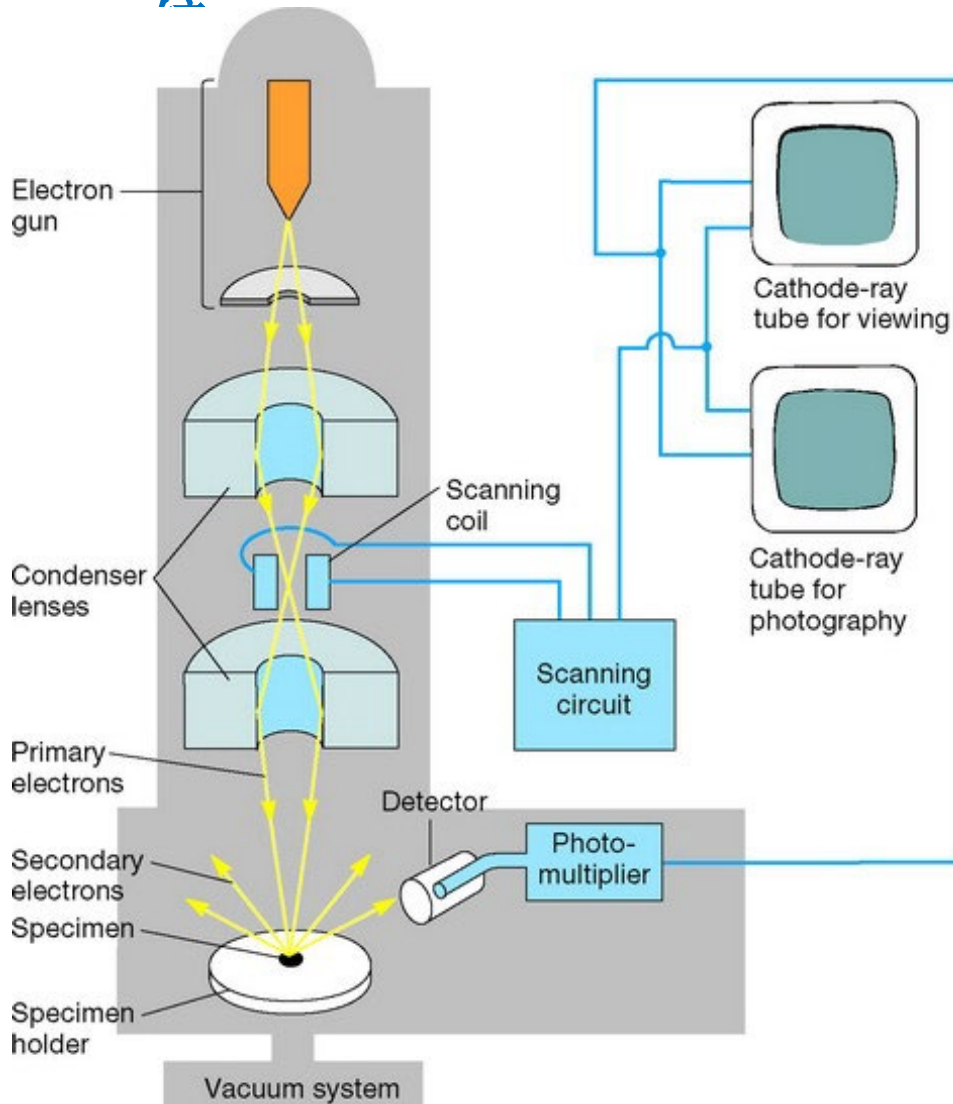
Excellent for flagella, etc



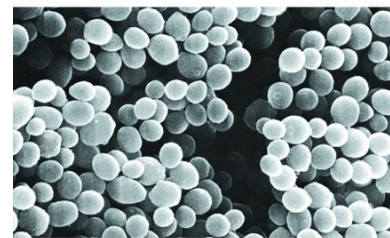
- a) Negatively stained T4, a virus that infects *E. coli*
 b) *Proteus mirabilis* (奇异变形杆菌) after shadowing
 ×42,750

Different types of Microscopes

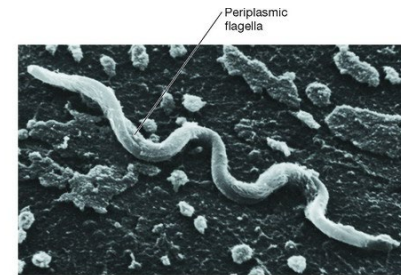
Scanning Electron Microscope (SEM) 扫描电子显微



- TEM forms an image from **radiation** that has **passed** through a specimen.
- SEM forms an image from **electrons released from** atoms on an object's surface.

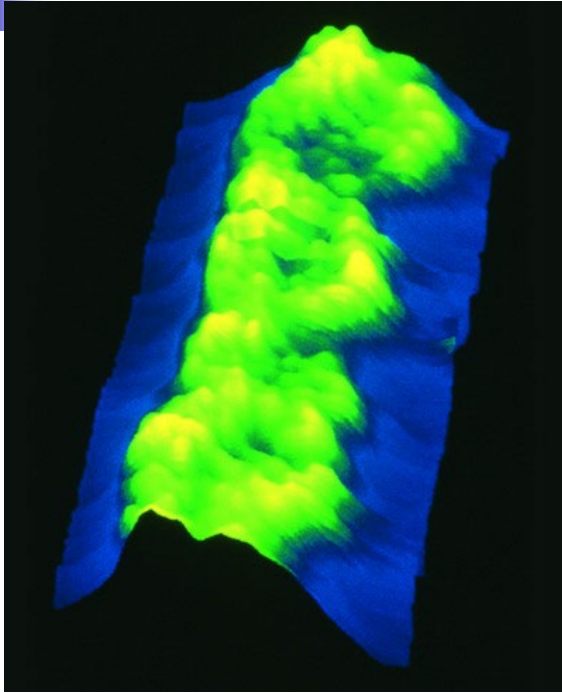


(a) *S. aureus*

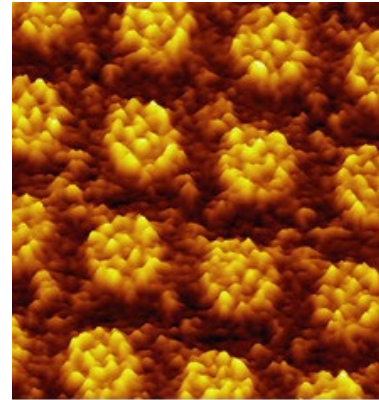


(b) *Cristispira*

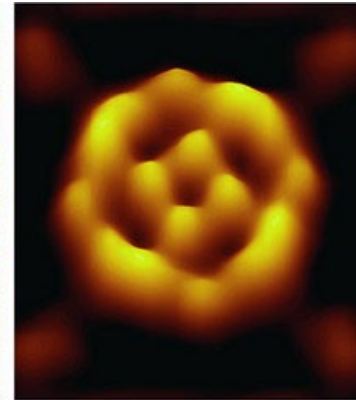
Scanning Probe Microscopy 扫描探针显微镜技术



Scanning Tunneling Microscopy of DNA
扫描隧道显微镜



(a)



(b)

The Membrane Protein Aquaporin
(水通道膜蛋白) Visualized by
Atomic Force Microscopy 原子力显
微镜 (a) The sureface view of a
single protein; (b) A single
aquaporin molecule in higher
magnification.

Comparisons of Types of Microscopy

Microscope	Max. Magnification	Resolution	Important Features
Visible light as source of illumination			
Bright-field	1,000×	0.2μm	For live and stained specimens; specimen is dark, field is white
Dark-field	2,000×	0.2μm	Best for observing live, unstained; Specimen is bright, field is black
Phase-contrast	2,000×	0.2μm	For live specimen, excellent for internal cellular detail
Differential interference	2,000×	0.2μm	Three-dimensional image of live specimens
Ultraviolet rays as source of illumination			
Fluorescent	2,000×	0.2μm	With fluorescent dyes, excellent diagnostic tool
Confocal Laser Scanning	2,000×	0.2μm	With fluorescent dyes, multiple images are combined into three dimensional image

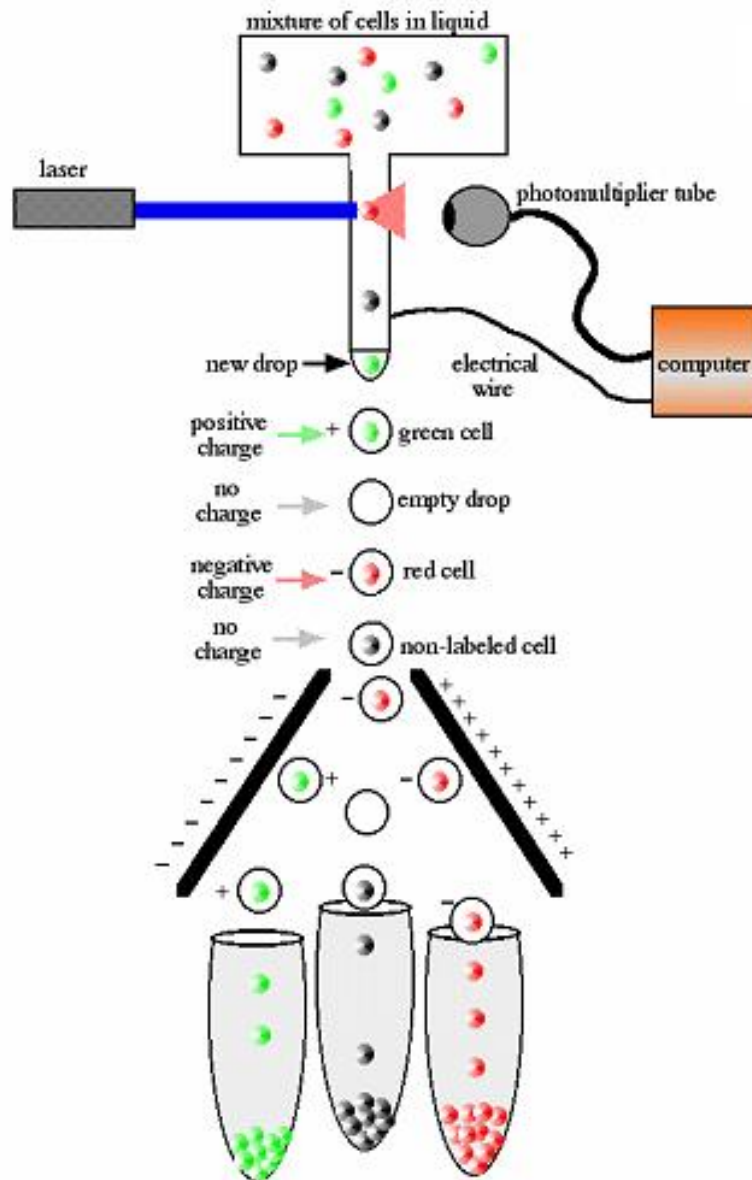


Microscope	Max. Magnification	Resolution	Important Features
Electron beam forms image of specimen			
Transmission electron microscope (TEM)	100,000×	0.2nm	Finest detailed structure of cells and viruses; used only on preserved material
Scanning electron microscope (SEM)	650,000×	10nm	For external surface of specimen; Three-dimensional image
Sharp tip probes atomic structure of specimen			
Atomic Force Microscope (AFM)	100,000,000×	0.01 Angstroms*	Tip moves and scans specimen, movement is measured by laser and translated to image. For protein, DNA
Scanning Tunneling Microscope (STM)	100,000,000×	0.01 Angstroms*	Tip moves and voltage is applied to measure the distance. For protein, DNA

* Angstroms, 埃= 10^{-10} m

Flow Cytometry

流式细胞仪

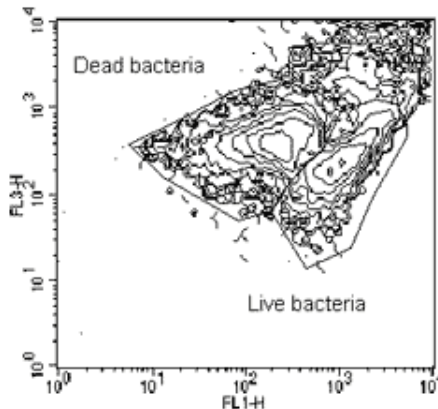
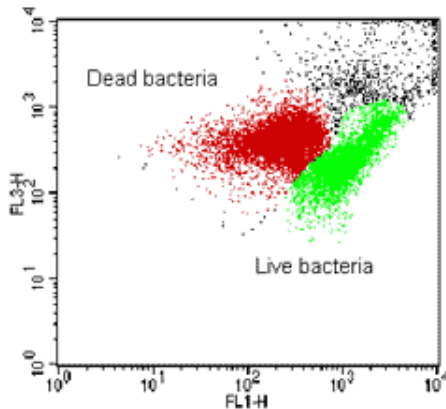


- Fluorescent cells are injected into a fluid stream that narrows, forcing only one cell at a time to pass a laser beam
- The cells scatter light in all directions
- The light scattering properties of each cell (size, fluorescence of specific stains, auto-fluorescence of chlorophylls and other pigments) distinguishes each cell type
- The detector software identifies the fluorescent signal: intensity depends on cell size and complexity
- Cells become charged as they pass the detector
- The particles are diverted from the stream according to charge, and collected (termed **Cell Sorting**)

细胞分选

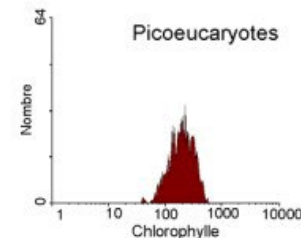
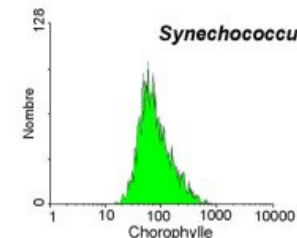
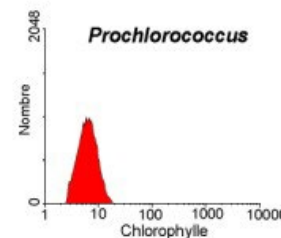
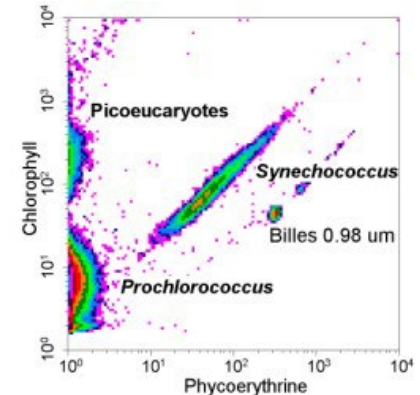
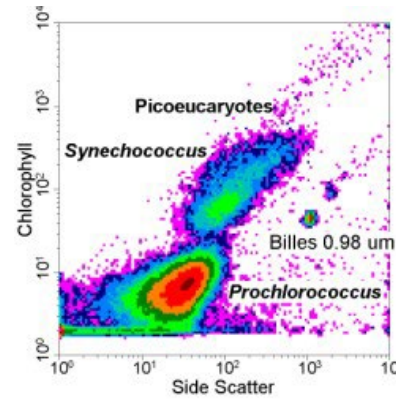
Flow Cytometry

- several 1000 cells per second can be counted
- The capacity to rapidly count large numbers of particles greatly increases the reliability and precision of cell concentration estimates
- However the cells are not directly identified and microscopy etc. may then be needed



Data from Antarctic Division analysis of live vs dead cells in Antarctic waters

<http://www.antarctica.gov.au/about-antarctica/wildlife/microscopic-organisms/marine-microbial-ecology/how-are-microbes-studied/flow-cytometry>



Analysis of a marine sample of photosynthetic picoplankton by flow cytometry showing three different populations

(*Prochlorococcus* 原绿球藻, *Synechococcus* 聚球藻, and picoeukaryotes 微微型真核藻)

(From Wikipedia)

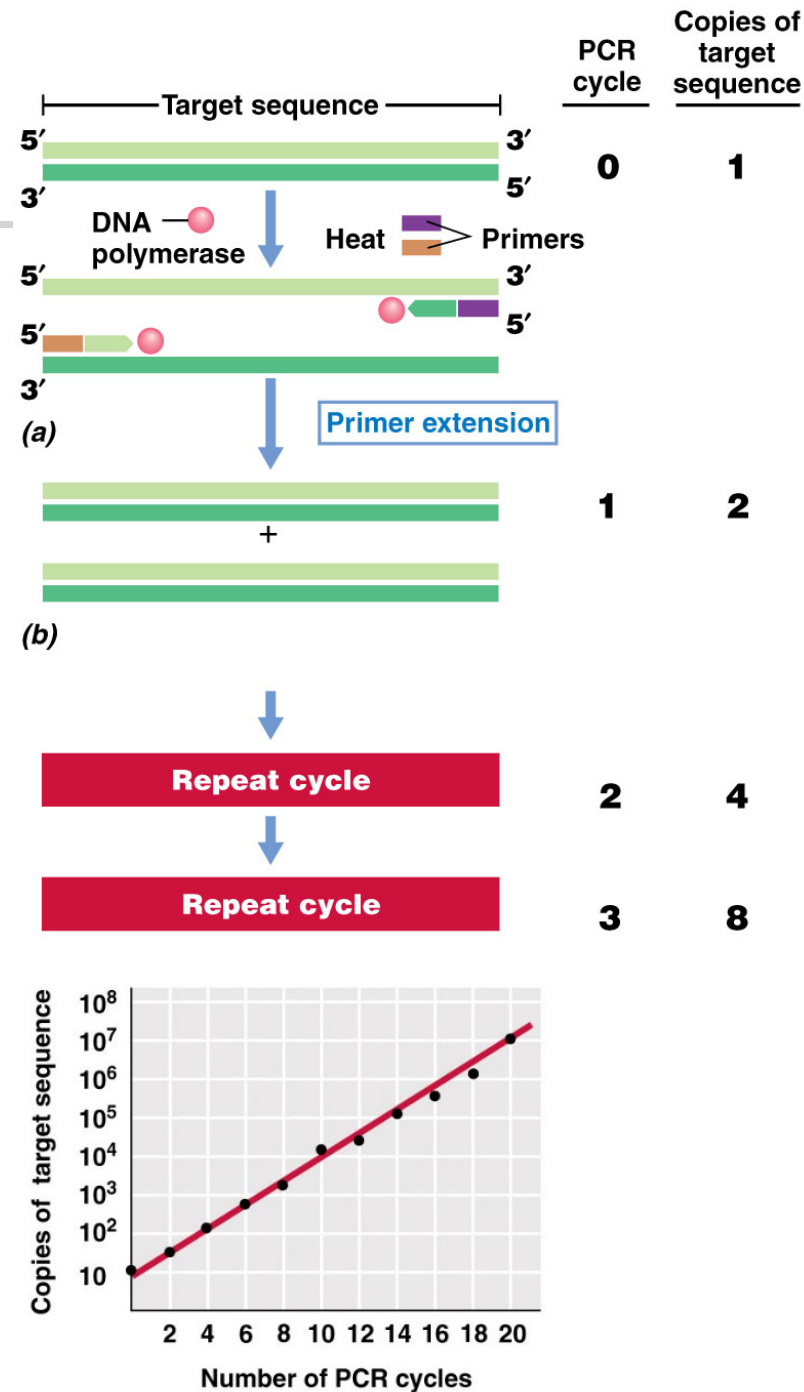
PCR-based methods for community analysis

The *polymerase chain reaction (PCR)*

- Discovery of the PCR has enabled detailed analysis of diversity in microbial communities in natural habitats

In the PCR:

- DNA is extracted from a sample (could be a cultured species or a natural sample)
- A specific gene sequence (the 'target gene/sequence') in the DNA is detected by 'primers' and *amplified* (many copies made) using the enzyme *DNA polymerase*





PCR (Polymerase Chain Reaction) 聚合酶链式反应

HHMI

保守序列

- A specific gene sequence has conserved sequence in the same genus (get conserved sequence from published papers).
- According to the conserved sequence, we design **the primers** (引物).

[illegible]

Primer forward

NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNAATGCCCTCGGGGCTC
|||||||
-----TTACGGGGAGCCCCGAG

Primer reverse

PCR

Reaction Buffer

Template

Primers

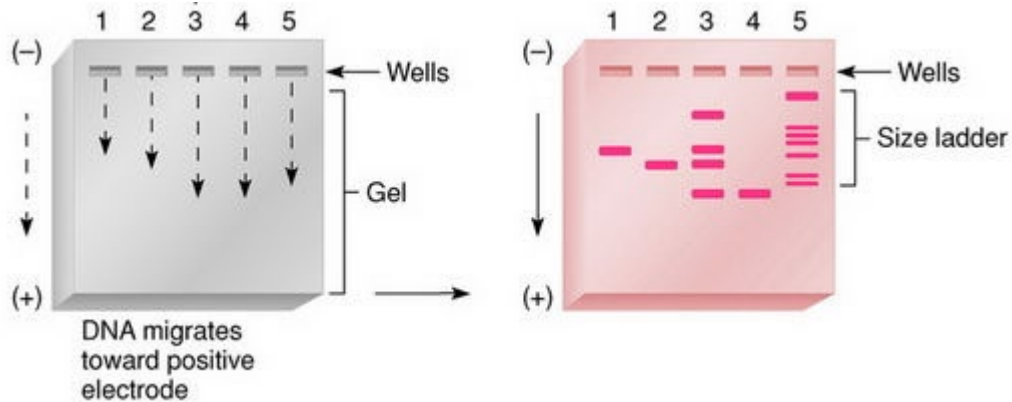
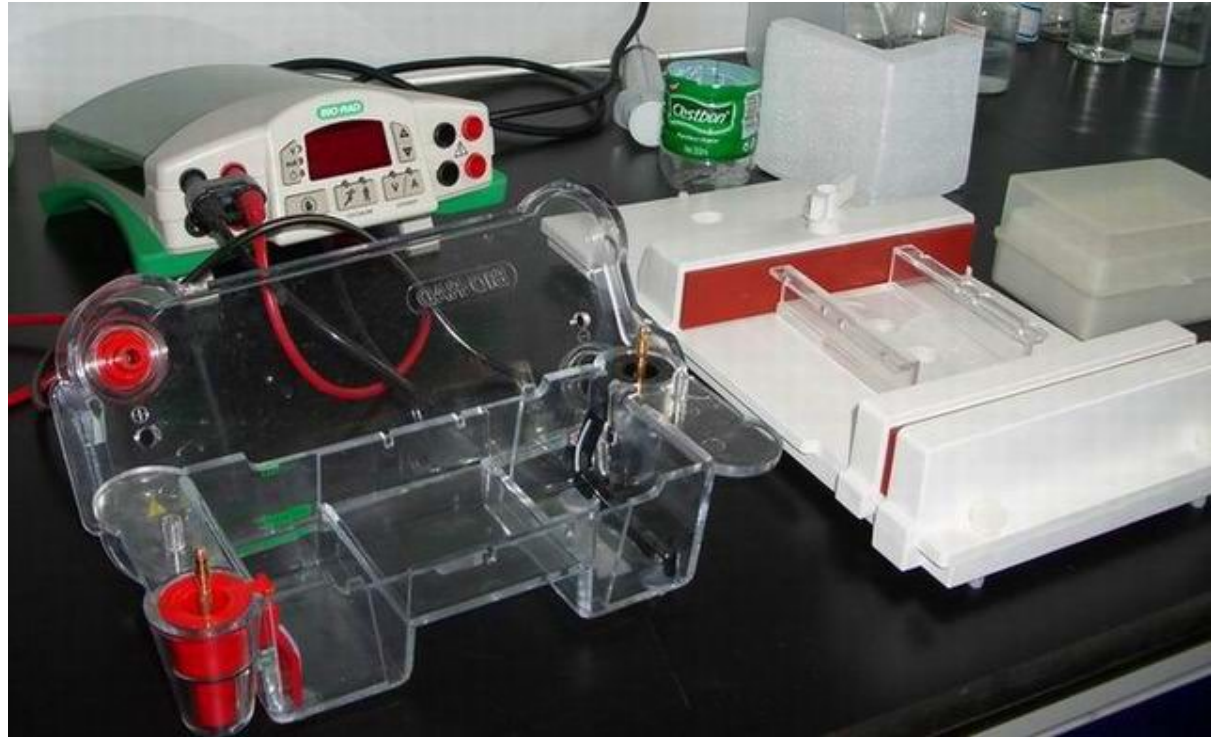
dNTPs (ATCG)

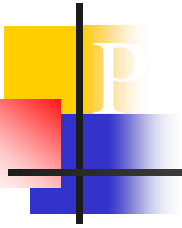
Polymerase(Taq)



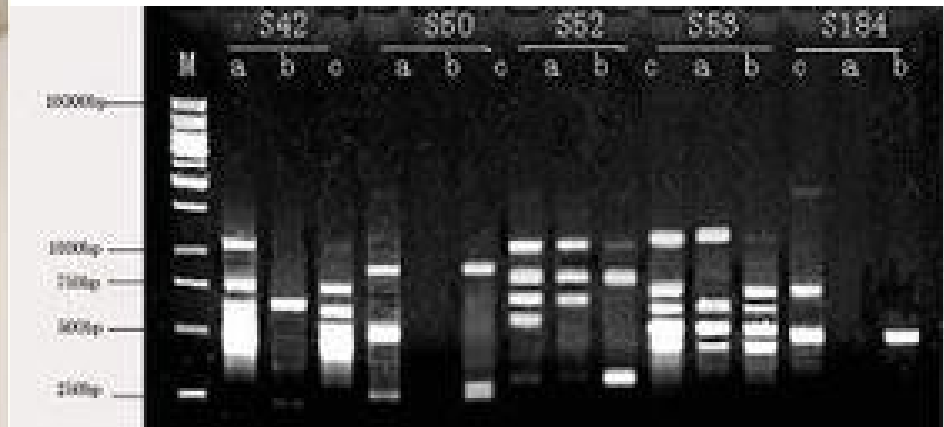
Detection after PCR——Gel Electrophoresis

凝胶电泳





Detection after PCR——Gel Electrophoresis



PCR-based methods for community analysis

The *polymerase chain reaction (PCR)*

• A specific gene sequence (the 'target gene/sequence') in the DNA is detected by 'primers' and *amplified* (many copies made) using the enzyme *DNA polymerase*

1. could be a gene useful in **phylogenetic (diversity) analysis** among taxa
 - e.g. 16S rRNA gene
 - The internal transcribed spacer region (ITS) between the 16S and 23S rRNA genes may also be used
2. could be a gene **conferring particular functions**
 - Some common genes shown in table – gene symbols and enzymes need not be memorized

Table 22.3 Genes commonly used for evaluating specific microbial processes in the environment using PCR

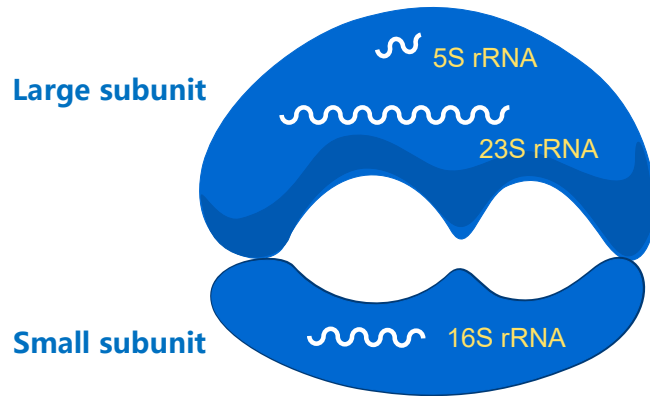
Metabolic process ^a	Target gene	Encoded enzyme
Denitrification	<i>narG</i>	Nitrate reductase
	<i>nirK, nirS</i>	Nitrite reductase
	<i>norB</i>	Nitric oxide reductase
	<i>nosZ</i>	Nitrous oxide reductase
Nitrogen fixation	<i>nifH</i>	Nitrogenase
Nitrification	<i>amoA</i>	Ammonia monooxygenase
Methane oxidation	<i>pmoA</i>	Methane monooxygenase
Sulfate reduction	<i>apsA</i>	Adenosine phosphosulfate reductase
	<i>dsrAB</i>	Sulfite reductase
Methane production	<i>mcrA</i>	Methyl coenzyme M reductase
Degradation of petroleum compounds	<i>nahA</i>	Naphthalene dioxygenase
	<i>alkB</i>	Alkane hydroxylase
Anoxygenic photosynthesis	<i>pufM</i>	M subunit of photosynthetic reaction center

Phylogenetic analysis

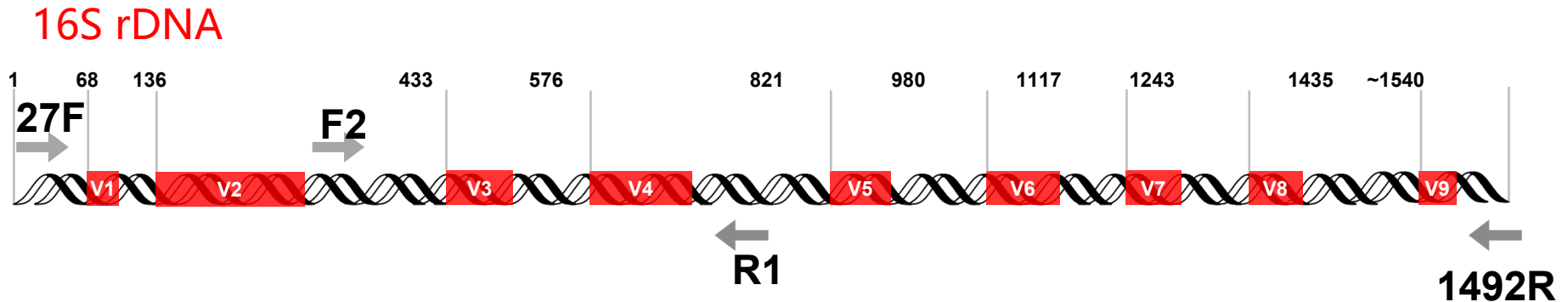
基于**16s/18s rDNA**的系统发育学分析

- rDNA——ribosomal DNA, 鉴定物种的标准标识序列
- The rDNAs have low rate of polymorphism among species, which allows interspecific comparison to elucidate phylogenetic relationship
- Identify bacteria——16S rDNA by PCR with universal primers 通用引物
- Identify Fungi——18S rDNA, 28S rDNA, ITS by PCR with universal primers

Prokaryotic ribosome (原核生物的核糖体)



- ✓ Prokaryotic RNA (ribosomal ribonucleic acid , rRNA) including 5S,16S and 23S rRNA。
- ✓ 16S rDNA coding for **16S rRNA**, exists in all **prokaryotes**.

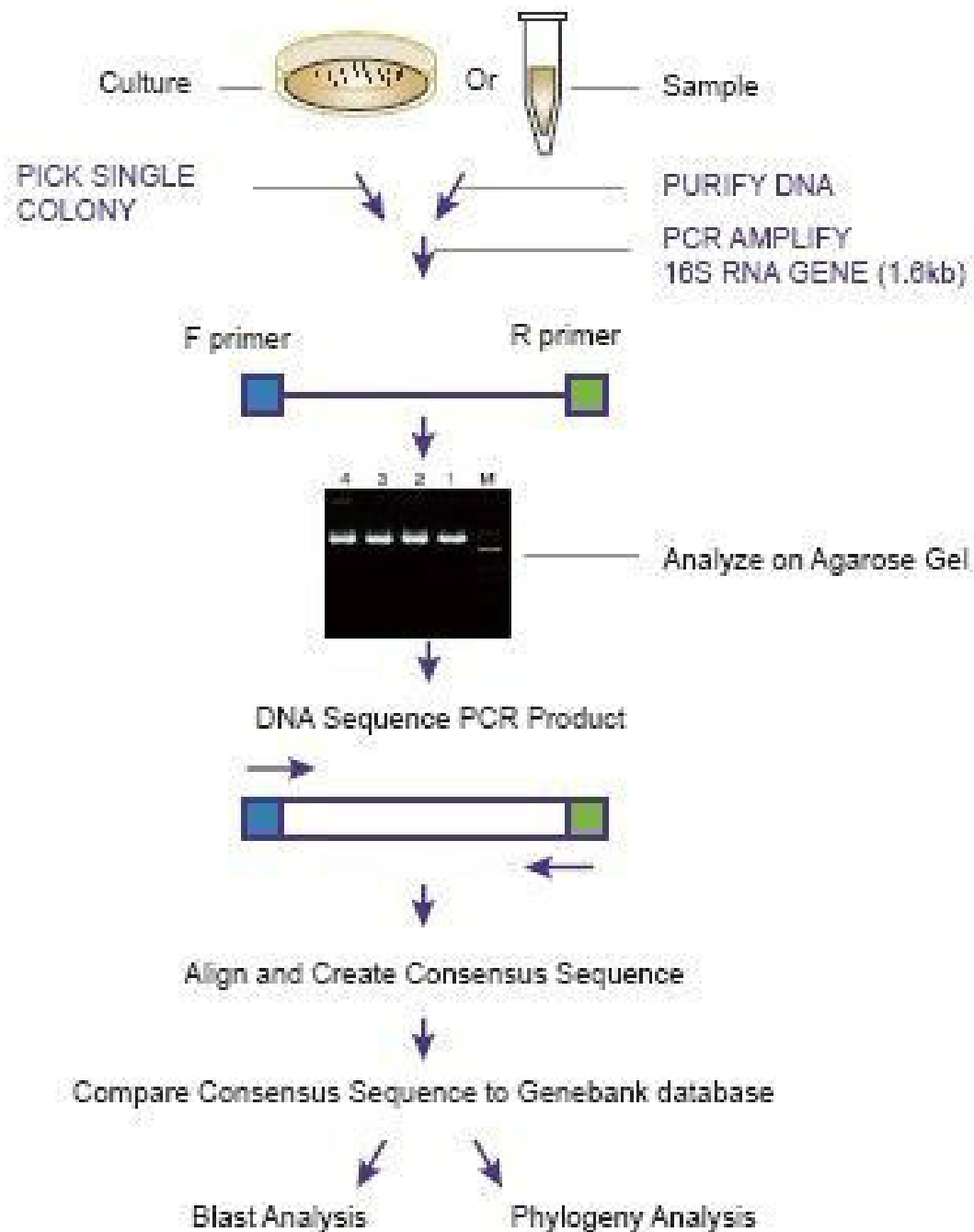
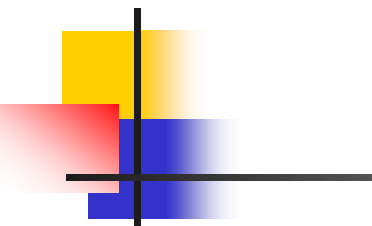


- ✓ 16S rDNA 1540bp, including 9 variable regions and 10 conserved regions.
- ✓ Primers designed according to the conserved regions.

Phylogenetic analysis

Procedure:

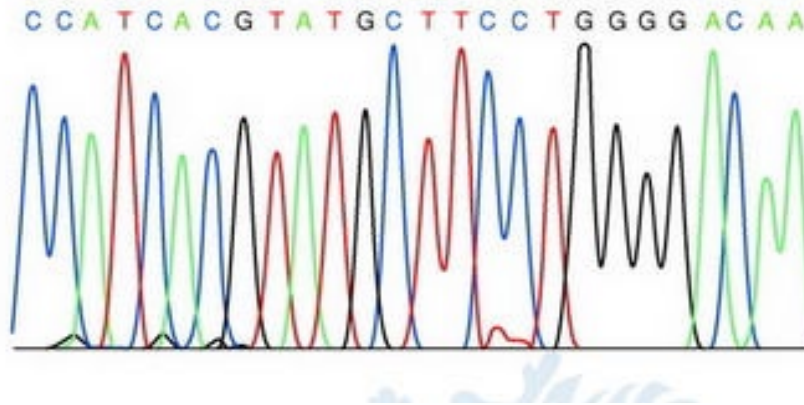
1. Extract genome DNA from strains
2. PCR with universal primers
3. Send PCR mixture or purified DNA to biological company for sequencing
4. Blast after we get sequence results from company



➤ DNA sequencing: Determining the Exact Genetic Code

*.ab1

(chromas program)



*.seq

(text)

>sequence 1

```
TATATATATATATATATACAGTGTTTGAACCTGGGGGACCCTATACCTGTAGTGTCTTGAGTGCT
TTCTAGTGTTTTTATCCCGACGTCTAATAGTCCATTTGGACTATTTTTGCATATATGTCAAAATATCT
GAGTAACAGGTACGTGGTTGTTAGTTGATAAATAGCGGCTAAAACGGCCCTGTTAACTTTGCACA
CCTCAACTATCAAGTTCTATACCTATCGACCCTCTTGGAACACGAGCCTATATAGAATACGCACGG
AATTTTCGTAGTTGAGCTGAATATATGTACTAGGAAATTAATCATACCTAAATGAGGAGAAAGAATA
TACTTTCAAGTACGAATAAAGCGCAATGGTTGGCAAACGCTGTAATGCACGTGGTACAGTGGTAA
GTTTGAATGTGAATATCAACATATGACACTGGATACATTCATA
```


BLASTn

NCBI (美国国立生物技术信息中心) 比对
<http://blast.ncbi.nlm.nih.gov/Blast.cgi>

BLAST® >> blastn suite

Standard Nucleotide BLAST

blastn blastp blastx tblastn tblastx

BLASTn programs search nucleotide databases using a nucleotide query. [more...](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#) Query subrange [?](#)

From

To

Or, upload file [浏览...](#) 未选择文件. [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database ☐ Standard databases (nr etc.) ☒ rRNA/ITS databases ☐ Genomic + transcript databases ☐ Betacoronavirus

Nucleotide collection (nr/nt) [?](#)

Limit by ☐ Organism ☐ BioProjectID ☐ WGS Project

Organism [Optional](#) ☐ exclude [+](#)

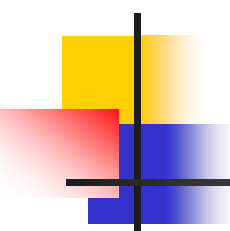
Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

Exclude [Optional](#) ☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to [Optional](#) ☐ Sequences from type material

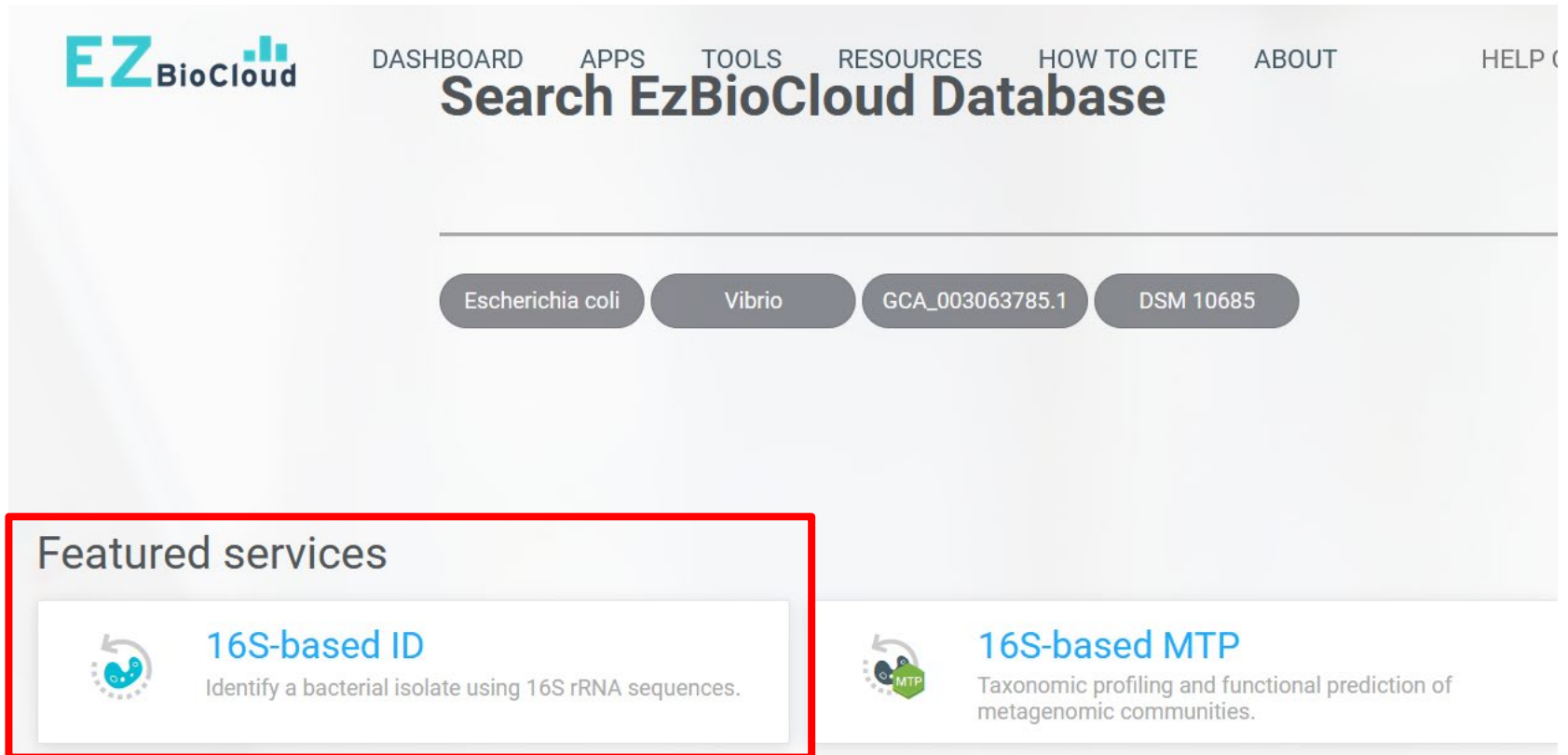
Entrez Query [Optional](#) [YouTube](#) [Create custom database](#)

Enter an Entrez query to limit search [?](#)



EZBioCloud

<https://www.ezbiocloud.net/>



The screenshot shows the EZBioCloud website. At the top left is the EZBioCloud logo. To its right is a navigation menu with links: DASHBOARD, APPS, TOOLS, RESOURCES, HOW TO CITE, ABOUT, and HELP. Below the navigation menu is a large search bar with the text "Search EzBioCloud Database". Under the search bar are four buttons: "Escherichia coli", "Vibrio", "GCA_003063785.1", and "DSM 10685". Below the search bar is a section titled "Featured services" which is highlighted with a red border. This section contains two service cards. The first card is titled "16S-based ID" and includes the text "Identify a bacterial isolate using 16S rRNA sequences." The second card is titled "16S-based MTP" and includes the text "Taxonomic profiling and functional prediction of metagenomic communities."

EZBioCloud

DASHBOARD APPS TOOLS RESOURCES HOW TO CITE ABOUT HELP

Search EzBioCloud Database

Escherichia coli Vibrio GCA_003063785.1 DSM 10685

Featured services

 **16S-based ID**
Identify a bacterial isolate using 16S rRNA sequences.

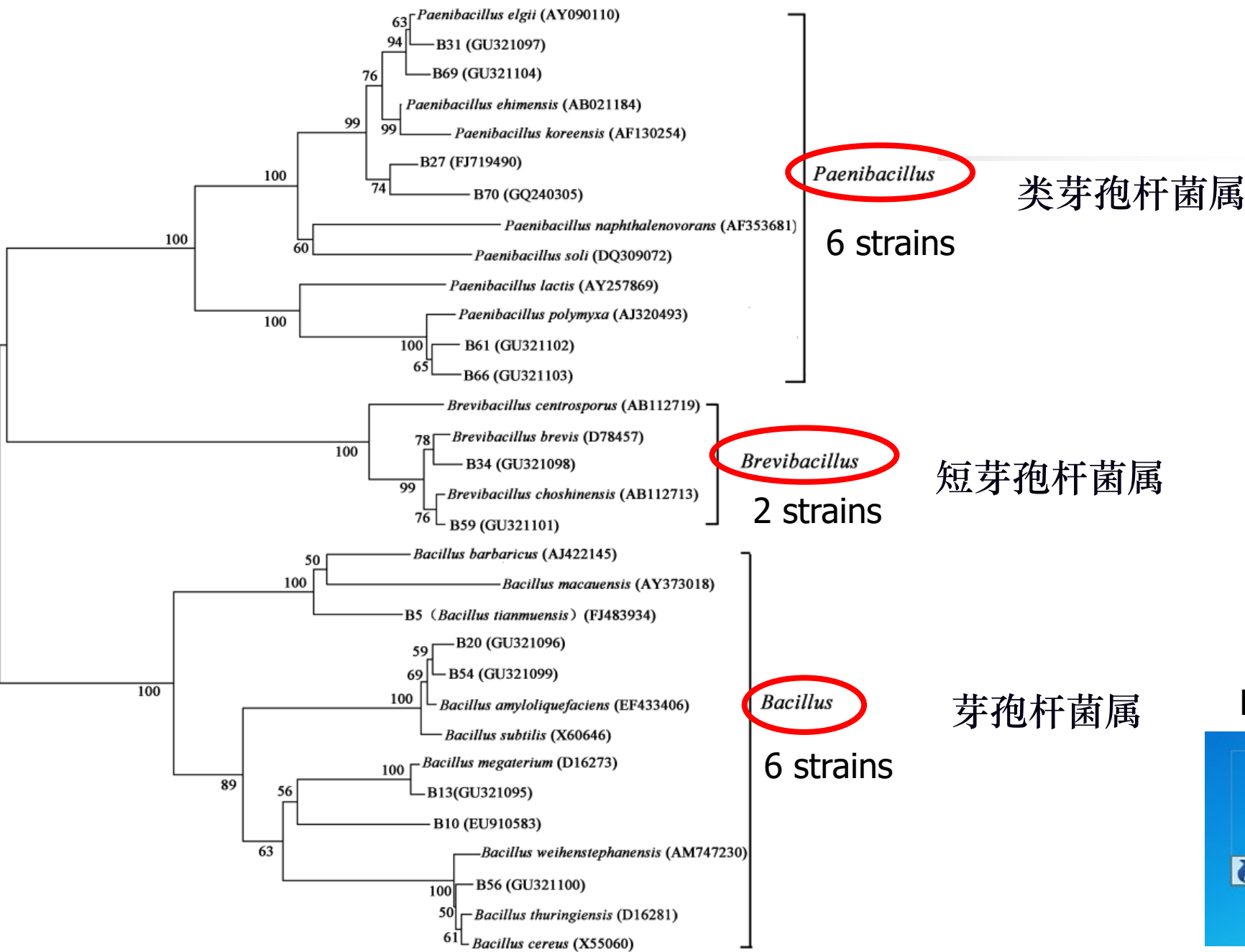
 **16S-based MTP**
Taxonomic profiling and functional prediction of metagenomic communities.

➤ For 16s rDNA (bacteria)

if max ident < 98.6 % , highly possible novel species

Accession	Description	Max score	Total score	Query coverage	▲ E value	Max ident
NW_001884680.1	Aspergillus oryzae RIB40 contig SC206	3051	3051	99%	0.0	99%
NW_001849579.1	Aspergillus oryzae RIB40 contig rDNA_te13	3051	3051	99%	0.0	99%
NT_166520.1	Aspergillus niger CBS 513.88 supercontig An03	3046	1.025e+04	99%	0.0	99%
NC_007197.1	Aspergillus fumigatus Af293 chromosome 4, whole genome shotgun project	3046	4439	99%	0.0	99%
NT_165929.1	Aspergillus terreus NIH2624 scaffold_6 genomic scaffold, whole genome	2972	2972	99%	0.0	98%
NW_001510400.1	Aspergillus clavatus NRRL 1 1099423829788 genomic scaffold, whole genome	2918	2918	94%	0.0	99%
NW_002990118.1	Talaromyces stipitatus ATCC 10500 scf_1105507295515, whole genome	2915	2915	99%	0.0	98%
NW_003101676.1	Ajellomyces dermatitidis SLH14081 genomic scaffold supercont1.7	2824	1.493e+04	99%	0.0	97%
NW_003217216.1	Paracoccidioides brasiliensis Pb01 supercont1.72 genomic scaffold	2813	9289	99%	0.0	97%
NW_003217281.1	Paracoccidioides brasiliensis Pb01 supercont1.7 genomic scaffold	2813	8440	99%	0.0	97%

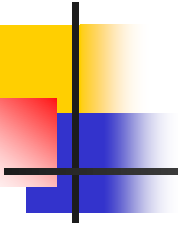
- Based on Blast results, we download the 16S rDNA of strains with similar sequences to construct [Phylogenetic tree](#) (usually by program [Mega](#))



Mega



MEGA7

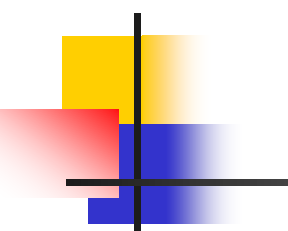


Metagenomics (environmental genomics)

Metagenomics is a molecular approach to taking a census of the entire microbial community

- It is the study of genomes recovered from environmental samples without first isolating and growing pure cultures of organisms
 - detects rare and low abundance groups; does not rely on PCR amplification
- Enables identification of the presence of many 'functional' genes and which genomic species (phylotypes) they are associated with
- These types of studies are beginning to unravel the microbial diversity of the oceans

Covered in subsequent lectures



Summary

- Marine microbiologists use a very wide variety of techniques to study the diversity and activity of microbes
- Long-established techniques such as culture and microscopy remain very important
- Most microbes cannot be grown in pure culture, novel approaches have been developed to allow isolation of previously uncultured species
- New molecular methods have led to the most significant advances in marine microbiology
- A survey of 16S/18S rDNA is the most common approach for assessing microbial diversity
- Microscopy/FISH and flow cytometry led to the recognition of the abundance and diversity of marine microbes